Null Space of A

The set $S = \{X \in \mathbb{R}^3 \mid AX = 0\}$ is known as null space of A.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 4 & 6 \end{pmatrix}$$

$$\approx \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \approx \begin{pmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

Rank (A) = 2

Nullity (A) = No. of Column − Rank (A)

= 3 − 2 = 1.

To find the null space of A, put

$$AX = 0$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \quad x_1 + 2x_2 + 3x_3 = 0$$

$$-2x_2 - 2x_3 = 0$$

$$\Rightarrow x_2 = -x_3$$

$$\therefore x_1 + 2(-x_3) + 3(x_3) = 0$$

$$\Rightarrow x_1 = -x_3$$

$$\therefore X = (-x_3, -x_3, x_3)$$

$$= x_3(-1, -1, 1)$$

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